

identify $M(x, y) = G(x, y)$ and $N(x, y) = -H(x, y)$ before using (4).

(ii) In some texts on differential equations the study of exact equations precedes that of linear DEs. If this were so, the method for finding integrating factors just discussed can be used to derive an integrating factor for $y' + P(x)y = f(x)$. By rewriting the last equation in the differential form $(P(x)y - f(x)) dx + dy = 0$ we see that

$$\frac{M_y - N_x}{N} = P(x).$$

From (13) we arrive at the already familiar integrating factor $e^{\int P(x) dx}$ used in Section 2.3.

2.4 Exercises

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–20, determine whether the given differential equation is exact. If it is exact, solve it.

1. $(2x - 1) dx + (3y + 7) dy = 0$
2. $(2x + y) dx - (x + 6y) dy = 0$
3. $(5x + 4y) dx + (4x - 8y^3) dy = 0$
4. $(\sin y - y \sin x) dx + (\cos x + x \cos y - y) dy = 0$
5. $(2xy^2 - 3) dx + (2x^2y + 4) dy = 0$
6. $\left(2y - \frac{1}{x} + \cos 3x\right) \frac{dy}{dx} + \frac{y}{x^2} - 4x^3 + 3y \sin 3x = 0$
7. $(x^2 - y^2) dx + (x^2 - 2xy) dy = 0$
8. $\left(1 + \ln x + \frac{y}{x}\right) dx = (1 - \ln x) dy$
9. $(x - y^3 + y^2 \sin x) dx = (3xy^2 + 2y \cos x) dy$
10. $(x^3 + y^3) dx + 3xy^2 dy = 0$
11. $(y \ln y - e^{-xy}) dx + \left(\frac{1}{y} + x \ln y\right) dy = 0$

12. $(3x^2y + e^y) dx + (x^3 + xe^y - 2y) dy = 0$
13. $x \frac{dy}{dx} = 2xe^x - y + 6x^2$
14. $\left(1 - \frac{3}{y} + x\right) \frac{dy}{dx} + y = \frac{3}{x} - 1$
15. $\left(x^2y^3 - \frac{1}{1 + 9x^2}\right) \frac{dx}{dy} + x^3y^2 = 0$
16. $(5y - 2x)y' - 2y = 0$
17. $(\tan x - \sin x \sin y) dx + \cos x \cos y dy = 0$
18. $(2y \sin x \cos x - y + 2y^2 e^{xy^2}) dx = (x - \sin^2 x - 4xye^{xy^2}) dy$
19. $(4t^3y - 15t^2 - y) dt + (t^4 + 3y^2 - t) dy = 0$
20. $\left(\frac{1}{t} + \frac{1}{t^2} - \frac{y}{t^2 + y^2}\right) dt + \left(ye^y + \frac{1}{t^2 + y^2}\right) dy = 0$

In Problems 21–26, solve the given initial-value problem.

21. $(x + y)^2 dx + (2xy + x^2 - 1) dy = 0, y(1) = 1$
22. $(e^x + y) dx + (2 + x + ye^y) dy = 0, y(0) = 1$
23. $(4y + 2t - 5) dt + (6y + 4t - 1) dy = 0, y(-1) = 2$
24. $\left(\frac{3y^2 - t^2}{y^5}\right) \frac{dy}{dt} + \frac{t}{2y^4} = 0, y(1) = 1$
25. $(y^2 \cos x - 3x^2y - 2x) dx + (2y \sin x - x^3 + \ln y) dy = 0, y(0) = e$
26. $\left(\frac{1}{1 + y^2} + \cos x - 2xy\right) \frac{dy}{dx} = y(y + \sin x), y(0) = 1$

In Problems 27 and 28, find the value of k so that the given differential equation is exact.

27. $(y^3 + kxy^4 - 2x) dx + (3xy^2 + 20x^2y^3) dy = 0$
28. $(6xy^3 + \cos y) dx + (2kx^2y^2 - x \sin y) dy = 0$

In Problems 29–32, verify that the given differential equation is not exact. Multiply the given differential equation by the indicated

integrating factor $\mu(x, y)$ and verify that the new equation is exact. Solve.

29. $(-xy \sin x + 2y \cos x) dx + 2x \cos x dy = 0; \quad \mu(x, y) = xy$
 30. $(x^2 + 2xy - y^2) dx + (y^2 + 2xy - x^2) dy = 0; \quad \mu(x, y) = (x + y)^{-2}$
 31. $(x^3y^2 + y) dx + x dy = 0; \quad \mu(x, y) = 1/x^2y^2$
 32. $x dx + (y - x^2 - y^2) dy = 0; \quad \mu(x, y) = 1/(x^2 + y^2)$

In Problems 33–38, solve the given differential equation by finding, as in Example 4, an appropriate integrating factor.

33. $(2y^2 + 3x) dx + 2xy dy = 0$
 34. $y(x + y + 1) dx + (x + 2y) dy = 0$
 35. $6xy dx + (4y + 9x^2) dy = 0$
 36. $\cos x dx + \left(1 + \frac{2}{y}\right) \sin x dy = 0$
 37. $(10 - 6y + e^{-3x}) dx - 2 dy = 0$
 38. $(y^2 + xy^3) dx + (5y^2 - xy + y^3 \sin y) dy = 0$

In Problems 39 and 40, solve the given initial-value problem by finding, as in Example 4, an appropriate integrating factor.

39. $x dx + (x^2y + 4y) dy = 0, y(4) = 0$
 40. $(x^2 + y^2 - 5) dx = (y + xy) dy, y(0) = 1$
 41. (a) Show that a one-parameter family of solutions of the equation

$$(4xy + 3x^2) dx + (2y + 2x^2) dy = 0$$

is $x^3 + 2x^2y + y^2 = c$.

(b) Show that the initial conditions $y(0) = -2$ and $y(1) = 1$ determine the same implicit solution.

(c) Find explicit solutions $y_1(x)$ and $y_2(x)$ of the differential equation in part (a) such that $y_1(0) = -2$ and $y_2(1) = 1$. Use a

graphing utility to graph $y_1(x)$ and $y_2(x)$.

Discussion Problems

- 42.** Consider the concept of an integrating factor used in Problems 29–40. Are the two equations

$$M dx + N dy = 0 \text{ and } \mu M dx + \mu N dy = 0$$

necessarily equivalent in the sense that a solution of one is also a solution of the other? Discuss.

- 43.** Reread Example 3 and then discuss why we can conclude that the interval of definition of the explicit solution of the IVP (the blue curve in Figure 2.4.1) is $(-1, 1)$.
- 44.** Discuss how the functions $M(x, y)$ and $N(x, y)$ can be found so that each differential equation is exact. Carry out your ideas.

(a) $M(x, y) dx + \left(x e^{xy} + 2xy + \frac{1}{x} \right) dy = 0$

(b) $\left(x^{-1/2} y^{1/2} + \frac{x}{x^2 + y} \right) dx + N(x, y) dy = 0$

- 45.** Differential equations are sometimes solved by having a clever idea. Here is a little exercise in cleverness: Although the differential equation

$$(x - \sqrt{x^2 + y^2}) dx + y dy = 0$$

is not exact, show how the rearrangement

$$\frac{x dx + y dy}{\sqrt{x^2 + y^2}} = dx$$

and the observation $\frac{1}{2} d(x^2 + y^2) = x dx + y dy$ can lead to a solution.

- 46.** True or False: Every separable first-order equation $dy/dx =$

$g(x)h(y)$ is exact.

Mathematical Model

- 47. Falling Chain** A portion of a uniform chain of length 8 ft is loosely coiled around a peg at the edge of a high horizontal platform and the remaining portion of the chain hangs at rest over the edge of the platform. See **FIGURE 2.4.2**. Suppose the length of the overhang is 3 ft, that the chain weighs 2 lb/ft, and that the positive direction is downward. Starting at $t = 0$ seconds, the weight of the overhanging portion causes the chain on the table to uncoil smoothly and fall to the floor. If $x(t)$ denotes the length of the chain overhanging the table at time $t > 0$, then $v = dx/dt$ is its velocity. When all resistive forces are ignored, it can be shown that a mathematical model relating v and x is

$$xv \frac{dv}{dx} + v^2 = 32x.$$

- (a)** Rewrite the model in differential form. Proceed as in Problems 33–38 and solve the DE by finding an appropriate integrating factor. Find an explicit solution $v(x)$.
- (b)** Determine the velocity with which the chain leaves the platform.

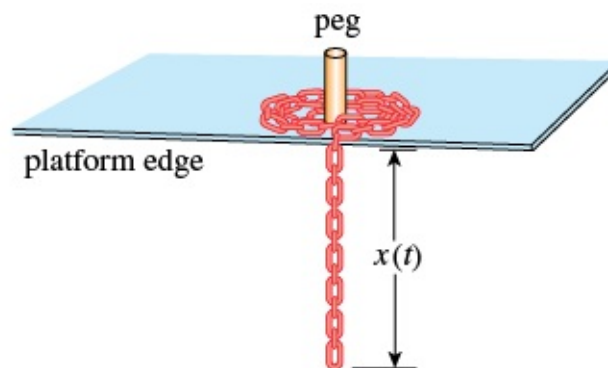


FIGURE 2.4.2 Uncoiling chain in Problem 47

Computer Lab Assignment

48. **Streamlines** (a) The solution of the differential equation

$$\frac{2xy}{(x^2 + y^2)^2} dx + \left[1 + \frac{y^2 - x^2}{(x^2 + y^2)^2} \right] dy = 0$$

is a family of curves that can be interpreted as streamlines of a fluid flow around a circular object whose boundary is described by the equation $x^2 + y^2 = 1$. Solve this DE and note the solution $f(x, y) = c$ for $c = 0$.

(b) Use a CAS to plot the streamlines for $c = 0, \pm 0.2, \pm 0.4, \pm 0.6$, and ± 0.8 in three different ways. First, use the **contourplot** of a CAS. Second, solve for x in terms of the variable y . Plot the resulting two functions of y for the given values of c , and then combine the graphs. Third, use the CAS to solve a cubic equation for y in terms of x .